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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION October 2026
Fifth SEMESTER
Discipline Specific Elective (DSE) - MG5DSEBOTA05 - BIO-INSTRUMENTATION AND QUALITY ANALYSIS 3

Duration: 2 Hours **Maximum Marks: 70**

Part A

Answer in One or Two Sentences Answer any 10 questions Each question carries 2 marks

1. **What are simple phenolics and polyphenolics?** [K] / [CO1]
Simple phenolics – containing single phenol group. They include simple phenols, phenolic acids and coumarins. Polyphenolics – containing more than one phenolic groups. They include flavonoids tannins and lignans.

2. **What is pseudoalkaloid?** [K] / [CO1]
Contain nitrogen but are not derived from amino acids. The core carbon skeleton and nitrogen come from different biosynthetic routes.

3. **What you mean by secondary structure of RNA?** [U] / [CO1]
Local folding patterns driven by internal base pairing. These shapes include hairpin loops, bulges, stems, and internal loops.

4. **What are polyunsaturated fatty acids? Give an example** [U] / [CO1]
Unsaturated fatty acids with more than one double bonds are called polyunsaturated fatty acids of PUFA. They are considered as essential fatty acids as the body cannot synthesize them. Linoleic acid, linolenic acid, arachidonic acid.

5. **Explain the extraction chamber of a Soxhlet apparatus** [U] / [CO2]
a wide tube with same width at the top while the bottom is closed. In this chamber the plant material is placed. The extraction chamber has 2 siphons attached. The Vapour distillation path arise from the base stem and connects to the top region of the extraction chamber. The small siphon tube is a U shaped tube that arise from the base of the extraction tube, runs up half of the extraction tube and then runs down as a U and then opens in to round bottom.

6. **What is partition coefficient? How K value of a phytochemical determine solubility** [A] / [CO2]
It equals the concentration of solute in the organic phase divided by its concentration in the aqueous phase. $K > 1$: The phytochemical is lipophilic; $K < 1$: The phytochemical is hydrophilic

7. **Explain Bromine water test for tannin detection** [U] / [CO3]
Take 2 ml of a given sample in a clean, dry test tube. Add 5 drops of Bromine water. Observe the appearance of buff or white precipitation in the test tubes.

8. **How can you quantify DNA using UV spectrophotometer** [A] / [CO3]
DNA strongly absorb UV radiation with a peak absorption at wavelength of 260 nm. An absorbance of 1.0 at A260 nm corresponds to 50µg ds DNA

9. **What is DNS reagent? How can this be used in detecting reducing sugar** [K] / [CO3]
2,5 Dinitrosalicylic acid. Reacts with reducing sugar to form 2 amino 5 nitro salicylic acid that gives orange to red colour, intensity of colour proportional to concentration of reducing sugar.

10. **Explain Oil Red O staining of lipids** [U] / [CO3]
It works on the principle of physical adsorption. Apply fresh Oil Red O Working Solution for 10–20 mins. Wash immediately with water or 60% isopropanol, mount using an aqueous mounting medium. Lipid Droplets - Appear intensely bright red or orange-red.

11 **What is the advantage of freeze drying over rotary evaporation?** [U] / [CO4]
Volatile compounds are not lost, heat labile compounds can be concentrated by lyophilization.

12 **List out applications of GC MS analysis** [A] / [CO4]
Detection, quantification and identification of volatile compounds in plant sample. Detection of pesticides in food and environmental samples, detection of specific metabolites associated with cancer and other diseases

[2x10 = 20]

Part B

Short Answer Type Questions, Answer any 8 questions, Each question carries 5 marks

13 **What are the functions of flavanoid?** [K] / [CO1]
Pigmentation in flowers and fruits to attract pollinators and seed-dispersing animals. Contribute to taste and fragrance of plant parts, helping with pollination. Act as a natural sunscreen by absorbing harmful UV radiation. Scavenge free radicals preventing cellular damage. Act as phytoalexins and deterrents against herbivores, pathogens. Signals soil bacteria to form symbiotic root nodules that fix essential nitrogen

14 **Write a note on primary metabolites** [K] / [CO1]
compound that is directly involved in the normal growth, development, and reproduction. They perform essential physiological functions and their absence typically results in immediate cell death or severe growth inhibition. They are essential for survival, produced during the growth phase and highly conserved

15 **Explain the biosynthesis of carbohydrate** [U] / [CO1]
Photosynthesis- capture sunlight through chlorophyll. The light-dependent reactions generate ATP and NADPH, which are then used in the Calvin cycle to fix atmospheric CO_2 into simple sugars. Gluconeogenesis: The process of synthesizing glucose from non-carbohydrate sources like lactate, pyruvate, glycerol, and certain amino acids. Glycogenesis - Once glucose is synthesized, animals store excess glucose as glycogen, plants convert these sugars into sucrose for transport, starch for storage and cellulose for structural support in cell walls.

16 **Explain manual grinding and knife-blade-hammer grinding** [U] / [CO2]
Manual- sample is manually crushed and ground using Mortar and Pestle. Often inert abrasives, such as washed sand glass beads are added. Knife-blade-hammer- bulk processing of dried plant biomass (barks, roots, stems, dried leaves). a) Blade/Cutting Mills: Use sharp rotating blades to chop fiber-rich materials (like stems and leaves) into uniform, coarse fragments. b) Blender – Blender can be used to grind both fresh and dried plant samples. c) Hammer Mills: Use heavy mechanical "hammers" hitting at high speeds to crush brittle, dense elements like dry roots and wood into fine powders.

17 **Explain cold maceration method of metabolite extraction** [U] / [CO2]
Grind or chop the plant material into a coarse powder. Place the material in a stoppered container and pour the solvent over it until completely submerged. Seal the container and let it stand for 3 to 7 days

at room temperature, shaking or stirring the mixture periodically. Separate the liquid extract (micelle) from the remaining solid residue (marc) by decanting or filtering. Evaporate the solvent to concentrate the final bioactive extract.

18 **Explain Foam test and hemolysis test for saponin detection** [U] / [CO3]

Foam Test- When an aqueous plant extract containing saponins is violently shaken, these molecules align at the air-water interface and traps air pockets, creating a dense, persistent column of foam. Take 2 ml of given sample in a test tube. Add 5 ml of distilled water, stopper the test tube and shake vigorously for 30 seconds. Keep the tube upright. A thick, compact, honeycomb-like foam layer measuring at least 1 cm to 2 cm in height forms instantly and persists without collapsing for 15–30 minutes indicates saponin.

Hemolysis test- When a saponin-rich plant extract comes into contact with RBCs the saponins bind tightly to the cholesterol, and pull the sterols out of the bilayer. This forms physical macro-pores in the lipid bilayer. Water rushes into the cell due to colloid-osmotic pressure, causing the erythrocytes to swell, burst, and release hemoglobin into the surrounding medium (hemolysis). Take 2 ml of given sample in a test tube, add 5 ml 10% blood solution. Gently shake and observe the tubes after 10 minutes. The sample tube containing saponin turns from an opaque, cloudy suspension into a perfectly transparent, clear dark-red solution.

19 **Explain the principle and procedure of Bradford's method of protein determination** [U] / [CO3]

The assay relies on the binding of a dye called Coomassie Brilliant Blue G-250 to proteins. Under highly acidic conditions, the dye is naturally in a doubly protonated red/brown state. When the dye binds to proteins, it converts into a stable blue, unprotonated form. This blue protein-dye complex shows maximum absorbance at 595 nm. The increase in absorbance at 595 nm is directly proportional to the amount of protein present. Add 1 mL of the sample to a test tube. To control tube add 1 ml distilled water. To both tubes, add 1 ml of Bradford reagent. Mix the contents and keep it aside for 10 minutes. Observe the colour change and take the reading using a spectrophotometer at 595 nm with control tube as blank

20 **Explain Hager's test and Mayer's test for the detection of alkaloids** [U] / [CO3]

Mayers Test- Nitrogen atoms in basic alkaloids form coordinate bonds or insoluble complexes with potassium-mercuric ions. This results in the formation of a cream-colored or pale yellow precipitate. Take 2 ml of a given sample in a clean, dry test tube. Add 2 ml of concentrated HCl to both the tubes and mix them. Add 5 drops of Mayer's Reagent. The emergence of a white precipitate or a distinct green coloration serves as a positive confirmation for the presence of alkaloid compounds.

Hager's Test- Picric acid acts as an organic acid that forms crystalline picrate salts with basic alkaloid molecules. Take 2 ml of a given sample in a clean, dry test tube. Add 6 drops of Hagers's Reagent. Mix well. The formation of an yellow precipitate indicate the presence of alkaloids

21 **What is solvent-solvent extraction? Explain co-current method.** [U] / [CO4]

the crude plant extract dissolved in a solvent is treated with another pure immiscible solvent. This helps to transfer the compound on interest or the impurities into the fresh immiscible solvent. In a co-current (or forward-flow) solvent-solvent extraction mechanism, the feed solution (containing the target metabolite) and the immiscible extracting solvent travel through the extraction system in the same direction. It is useful for fast, equilibrium-limited, or reactive extractions where prolonged contact is undesirable. The mechanism relies on a repeating cycle of stages. Each stage consists of two distinct physical phases: Mixing (Mass Transfer) and Settling (Phase Separation).

22 **Explain IR spectrophotometer** [U] / [CO4]

an analytical instrument that measures how a chemical sample absorbs infrared light to identify functional groups and determine molecular structures. Infrared Source: Generates broad-spectrum IR radiation using heated elements like a Nernst glower, or nichrome coils. Sample Compartment: Holds solid, liquid, or gas cells. Monochromator - Dispersive instruments use prisms or diffraction grating.

Fourier transform IR (FTIR) instruments use a Michelson interferometer. Detector: Converts optical signals into electrical signals using thermal. Recorder / Processor: Plots transmittance or absorbance (y-axis) against wavenumber or wavelength (x-axis)

[5x8 = 40]

Part C

Essay Type Questions, Answer any 1 questions, Each question carries 10 marks

23 **Write an essay on Ultrasound assisted extraction and Super Critical Fluid extraction** [K] / [CO2]

Ultrasound assisted extraction- Ultrasound-assisted extraction (UAE) is a green, non-thermal technique that uses high-frequency, high-intensity sound waves to recover valuable secondary plant metabolites (such as phenolics, flavonoids, alkaloids, and essential oils) from plant matrices. Principle- Acoustic Cavitation. Bubble Formation: Low-frequency ultrasound waves (typically 20 kHz to several MHz) pass through a liquid solvent containing immersed plant material. Cavitation: The pressure cycles create microscopic vacuum bubbles that grow and violently collapse. Cell Disruption: The collapse generates localized high temperatures, high pressures, and intense shear forces. These forces break plant cell walls and membranes, accelerating the release of intracellular metabolites into the solvent

Super Critical Fluid extraction- separation technique that uses a fluid in a supercritical state (meaning it is kept at a temperature and pressure above its critical point where liquid and gas phases merge) —to dissolve and isolate specific target compounds from a solid or liquid matrix. A gas like carbon dioxide (CO_2) is pressurized and heated past its critical point, giving it gas-like diffusion properties and liquid-like dissolving power (Supercritical fluid). The supercritical fluid flows through a high-pressure vessel holding the raw material to dissolve the desired components. The system reduces pressure in a separator vessel, turning the CO_2 back into a gas so it releases the extracted product for collection.

24 **Explain vertical gel electrophoresis** [U] / [CO4]

Separation of charged molecules like proteins and short nucleic acid through a vertically placed gel. Gel made of Poly acrylamide (Acrylamide, bis acrylamides, APS and TEMED). Two gels- Resolving gel and stacking gel. Stacking gel (highly porous layer with a pH of 6.8) stack the proteins and resolving gel (dense layer with a higher pH of 8.8) separates protein based on charge and mass. Native PAGE- separation based on charge and mass, SDS- separation based on mass only- SDS non-reductive and SDS reductive PAGE. Stacking gel low voltage, for resolving gel, enhance the voltage. Buffer system TAE or TBE buffer. Once tracking dye reaches the end, electrophoresis is stopped and the gel is stained and viewed.

[10x1 = 10]